



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO3 ASSESSMENT TOOL 1

Name : S.Deepadharshan

Reg No : 192472046

Course Name : Ethical Hacking

Course Code : ITA1408

Submitted to : Dr Parthiban S

Questions :

1. Perform Nmap enumeration on scanme.nmap.org using host discovery, port scanning, service/version detection, OS fingerprinting and default scripts. Analyze the services, attack surface, risks and security recommendations.
2. Perform comprehensive Nmap enumeration of 192.168.56.101 (Metasploitable 2). Identify open ports, services, versions and OS information. Classify services as High, Medium or Low risk and discuss possible attack vectors.
3. Enumerate 192.168.56.102 (Windows workstation) using Nmap. Identify ports, file-sharing and remote-management services and OS details. Analyze least privilege, attack risks and recommend security hardening measures.
4. Perform detailed Nmap enumeration of 192.168.56.103 (OWASP Juice Shop). Identify ports, services, versions, OS and web services. Analyze web-service risks and recommend measures to reduce exposure.
5. Perform comprehensive Nmap enumeration of 192.168.56.1 (network gateway). Identify exposed management interfaces and services such as SSH, Telnet, SNMP, HTTP and HTTPS. Analyze security risks and recommend controls to secure the gateway.

QUESTION 1 – scanme.nmap.org

Objective

To perform network enumeration of scanme.nmap.org using Nmap and identify active host status, open ports, services, service versions and operating-system information.

Commands

`nmap -sn scanme.nmap.org`

```
(kali@kali)-[~]
└─$ nmap -sn scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 17:44 IST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.32s latency).
Other addresses for scanme.nmap.org (not scanned):
2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 1.02 seconds
```

`sudo nmap -A -T4 scanme.nmap.org`

```
(kali@kali)-[~]
└─$ sudo nmap -A -T4 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 17:45 IST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.31s latency).
Not shown: 993 filtered tcp ports
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  Elite
Device type: general purpose
Running: Linux 2.6.X|3.X
OS CPE: cpe:/o:linux:linux_kernel
OS details: Linux 2.6.32 - 3.10
Network Distance: 17 hops
Service Info: Host: scanme.nmap.org; OS: Linux; CPE: cpe:/o:linux:linux_kernel
TRACEROUTE
HOP RTT ADDRESS
1 0.31 ms scanme.nmap.org (45.33.32.156)
Nmap done: 1 IP address (1 host up) scanned in 14.30 seconds
```

Findings

- Target: scanme.nmap.org
- IP Address: 45.33.32.156
- Host Status: Up
- Open Ports: 21, 22, 80, 9929, 31337
- Services: FTP, SSH, HTTP, Nping Echo and Elite/tcpwrapped
- SSH Version: OpenSSH 6.6.1p1 Ubuntu
- HTTP Version: Apache 2.4.7 Ubuntu
- OS Detection: Linux-related identification; Nmap fingerprinting was not fully conclusive.

Risk Analysis

Exposed services increase the attack surface. An attacker can use port and service information to identify technologies, research vulnerabilities, attempt unauthorized access and perform further reconnaissance.

Recommendations

1. Disable unnecessary services.
2. Restrict SSH access.
3. Keep services updated.
4. Use firewall rules.
5. Monitor suspicious network activity.

Conclusion

Nmap successfully identified the externally accessible services of the target. The information can help security teams reduce unnecessary exposure and improve server security.

QUESTION 2 – Metasploitable 2

Objective

To enumerate the deliberately vulnerable Linux server 192.168.56.101 and identify its open ports, services, versions and operating-system information.

Commands

nmap -sn 192.168.56.101

```
(kali@kali)~$ nmap -sn 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 18:01 IST
Nmap scan report for 192.168.56.101
Host is up (0.00045s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
```

sudo nmap -A -T4 192.168.56.101

```
(kali@kali)~$ sudo nmap -A -T4 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 18:01 IST
Nmap scan report for 192.168.56.101
Host is up (0.00041s latency).
Not shown: 977 closed tcp ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind  2 (RPC #10000)
139/tcp   open  netbios-ssn Samba smbd 3.X (workgroup: WORKGROUP)
449/tcp   open  netbios-ssn Samba smbd 3.X (workgroup: WORKGROUP)
512/tcp   open  exec     netkit-rsh rexecd
513/tcp   open  login?   
514/tcp   open  shell    Netkit rshd
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
TRACEROUTE
HOP RTT ADDRESS
1 0.41 ms 192.168.56.101
Nmap done: 1 IP address (1 host up) scanned in 9.87 seconds
```

Findings

- Host Status: Up
- Port 21: FTP
- Port 22: SSH

- Port 23: Telnet
- Port 25: SMTP
- Port 80: HTTP
- Port 139: NetBIOS
- Port 445: SMB
- Port 3306: MySQL
- Port 5432: PostgreSQL
- OS: Linux

Risk Classification

High Risk: Services that provide remote access or are known to be vulnerable.

Medium Risk: Services exposed to the network but requiring additional conditions for exploitation.

Low Risk: Services with limited functionality or restricted exposure.

Attack Vectors

The enumeration information can help a penetration tester identify vulnerable services, select appropriate testing methods and prioritize potential attack vectors.

Recommendations

1. Disable unnecessary services.
2. Patch vulnerable software.
3. Restrict remote-access services.
4. Configure firewall rules.
5. Use strong authentication.

QUESTION 3 – Windows Workstation

Objective

To enumerate 192.168.56.102 and identify active ports, file-sharing services, remote-management services and operating-system information.

Commands

`nmap -sn 192.168.56.102`

```
(kali@kali)~$ nmap -sn 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 18:12 IST
Nmap scan report for 192.168.56.102
Host is up (0.00037s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

`sudo nmap -A -T4 192.168.56.102`

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 18:12 IST
Nmap scan report for 192.168.56.102
Host is up (0.00045s latency).
Not shown: 992 closed tcp ports
PORT      STATE SERVICE          VERSION
135/tcp   open  mrpc             Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds     Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp   open  ms-wbt-server    Microsoft Terminal Services
Device type: general purpose
Running: Microsoft Windows 7|8|10
OS CPE: cpe:/o:microsoft:windows7:sp1 cpe:/o:microsoft:windows_8cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 7 SP1 or 8 or 10
Network Distance: 1 hop
TRACEROUTE
HOP RTT     ADDRESS
1 0.32 ms  192.168.56.102
```

Findings

- Host Status: Up
- Port 22: SSH – OpenSSH
- Port 80: HTTP – Node.js/Express
- Port 3000: HTTP – Node.js Express
- OS: Linux

Security Analysis

Unnecessary file-sharing and remote-management services increase the attack surface. Exposed services may provide information useful for reconnaissance and could potentially support lateral movement if security controls are weak.

Recommendations

1. Disable unnecessary services.
2. Restrict SMB/file sharing.
3. Apply least privilege.
4. Use firewall restrictions.
5. Enable strong authentication and MFA where applicable.

QUESTION 4 – OWASP Juice Shop

Objective

To enumerate the OWASP Juice Shop server at 192.168.56.103 and identify its exposed web and supporting services.

Commands

```
nmap -sn 192.168.56.103
```

```
$ nmap -sn 192.168.56.103
Starting Nmap 7.94SVN( https://nmap.org ) at 2026-08-10 18:22 IST
Nmap scan report for 192.168.56.103
Host is up (0.00048s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
```

```
sudo nmap -A -T4 192.168.56.103
```

```

$ sudo nmap -A -T4 192.168.56.103
Starting Nmap 7.94SVN( https://nmap.org ) at 2026-08-10 18:30 IST
Nmap scan report for 192.168.56.103
Host is up (0.00046s latency).
Not shown: 997 closed tcp ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.4 (protocol 2.0)
80/tcp    open  http      Node.js Express framework
|_http-title: OWASP Juice Shop
|_http-server-header: Express
3000/tcp   open  http      Node.js (Express middleware)
5000/tcp   open  http      Node.js Express framework
Aggressive OS guesses: Linux 3.2 - 4.9 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1 0.41 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds

```

Findings

- Target: 192.168.56.103
- Host Status: **[RESULT]**
- Open Ports: **[RESULT]**
- Web Service: **[RESULT]**
- Service Versions: **[RESULT]**
- OS: **[RESULT]**

Risk Analysis

Exposed web services increase the attack surface of the application. Service and version information can help identify technologies and potential vulnerabilities for further authorized testing.

Recommendations

1. Restrict unnecessary services.
2. Keep the web server updated.
3. Use firewall controls.
4. Secure administrative interfaces.
5. Perform regular vulnerability assessments.

QUESTION 5 – Network Gateway

Objective

To enumerate the network gateway at 192.168.56.1 and identify exposed management interfaces and network services.

Commands

`nmap -sn 192.168.56.1`

```
$ nmap -sn 192.168.56.1
Starting Nmap 7.94SVN( https://nmap.org ) at 2026-08-10 18:30 IST
Nmap scan report for 192.168.56.1
Host is up (0.00028s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

`sudo nmap -A -T4 192.168.56.1`

```
$ sudo nmap -A -T4 192.168.56.1
Starting Nmap 7.94SVN( https://nmap.org ) at 2026-08-10 18:30 IST
Nmap scan report for 192.168.56.1
Host is up (0.00025s latency).
Not shown: 995 closed tcp ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
|_http-title: Site doesn't have a title (text/html).
443/tcp   open  ssl/http Apache httpd 2.4.18 ((Ubuntu))
|_http-title: 400 Bad Request
|_ssl-cert: Subject: commonName=localhost
|_Not valid before: 2016-04-11T10:54:07
|_Not valid after: 2026-04-08T10:54:07
MAC Address: 08:00:27:02:22:8A (Oracle VirtualBox virtual NIC)
Aggressive OS guesses: Linux 3.13 - 4.4 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.25 ms 192.168.56.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 5.23 seconds
```

Findings

- Host Status: Up
- Port 22: SSH
- Port 80: HTTP
- Port 443: HTTPS
- OS: Linux

Security Analysis

Exposed management services such as SSH, Telnet, SNMP, HTTP or HTTPS can increase the risk of unauthorized access if they are poorly configured.

Recommendations

1. Disable Telnet.
2. Restrict management interfaces to trusted networks.
3. Use SSH instead of insecure protocols.
4. Change default credentials.
5. Use strong authentication.
6. Apply firewall rules.
7. Monitor management access.

Conclusion

Network enumeration helps identify unnecessary services and exposed management interfaces. Restricting these services and applying least-privilege principles can reduce the gateway's attack surface.